Econometrics Midterm · Short Answer

1. **[Part A: Econometrics 1–3 (Professor's Slides)]** What is the difference between a population and a sample?

2. **[Part A: Econometrics 1–3 (Professor's Slides)]** Why do econometricians usually study a sample instead of the population, and what is the goal of studying the sample?

3. **[Part A: Econometrics 1–3 (Professor's Slides)]** Define external validity and state which property of the sample matters for it.

4. **[Part A: Econometrics 1–3 (Professor's Slides)]** In $\log(\text{price}) = \beta_0 + \beta_1 \text{subsidy} + \beta_2 \log(\text{sunlight}) + u$, why is β_1 called a partial slope rather than simply "the slope"?

5. **[Part A: Econometrics 1–3 (Professor's Slides)]** In $\log(\text{price}) = \beta_0 + \beta_1 \text{battery} + \beta_2 \text{sunlight} + \beta_3 \text{battery}\times\text{sunlight} + u$ with $\beta_3 \neq 0$, why does battery not have a linear relationship with log(price)?

6. **[Part A: Econometrics 1–3 (Professor's Slides)]** What is the difference between random sampling and random assignment?

7. **[Part A: Econometrics 1–3 (Professor's Slides)]** For Model 2, $\text{wage} = \beta_0 + \beta_1 \text{educ} + u$ (ability not included), explain why the zero conditional mean condition may be violated.

8. **[Part A: Econometrics 1–3 (Professor's Slides)]** What is a control variable, and why is ability added in Model 1, $\text{wage} = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{abil} + u$?

9. **[Part B: Ch.1 The Nature of Econometrics and Economic Data]** What does "ceteris paribus" mean, and why is it important in econometrics?

10. **[Part B: Ch.1 The Nature of Econometrics and Economic Data]** What is panel (longitudinal) data, and what is one advantage of it?

11. **[Part C: Ch.2 The Simple Regression Model]** What does OLS minimize, and what is a residual?

12. **[Part C: Ch.2 The Simple Regression Model]** What does R-squared measure in a regression?

13. **[Part C: Ch.2 The Simple Regression Model]** What does the homoskedasticity assumption (SLR.5) state?

14. **[Part C: Ch.2 The Simple Regression Model]** What does it mean that the OLS estimator $\hat{\beta}_1$ is unbiased?

15. **[Part D: Ch.3 Multiple Regression Analysis: Estimation]** When does omitting a relevant-looking variable x_2 NOT cause bias in the estimated coefficient on x_1?

16. **[Part D: Ch.3 Multiple Regression Analysis: Estimation]** What is the main consequence of multicollinearity (high but not perfect correlation among regressors) for OLS?

17. **[Part D: Ch.3 Multiple Regression Analysis: Estimation]** State the Gauss–Markov theorem.

18. **[Part E: Ch.4 Multiple Regression Analysis: Inference]** Define the p-value of a hypothesis test.

19. **[Part E: Ch.4 Multiple Regression Analysis: Inference]** Does a statistically significant coefficient always matter in practice? Answer in 1–2 sentences.

20. **[Part E: Ch.4 Multiple Regression Analysis: Inference]** Interpret a 95% confidence interval for β_j.

21. **[Part E: Ch.4 Multiple Regression Analysis: Inference]** Which can make the usual OLS t statistics invalid (no t distribution under H_0): (i) heteroskedasticity, (ii) a sample correlation of 0.95 between two regressors in the model, (iii) omitting an important explanatory variable? Explain briefly.

22. **[Part F: Ch.5 Multiple Regression Analysis: OLS Asymptotics]** What does it mean that an estimator is consistent, and which assumptions are enough for OLS to be consistent?

23. **[Part F: Ch.5 Multiple Regression Analysis: OLS Asymptotics]** pctstck = $\beta_0 + \beta_1$ funds + β_2 risktol + u satisfies MLR.1–MLR.4 (risktol = risk tolerance). If funds and risktol are positively correlated and pctstck is regressed on funds only, what is the sign of the inconsistency in $\tilde{\beta}_1$, and why?

24. **[Part F: Ch.5 Multiple Regression Analysis: OLS Asymptotics]** Can cigs (average cigarettes smoked per day) have a normal distribution in the adult population? What does this imply for OLS inference?

25. **[Part F: Ch.5 Multiple Regression Analysis: OLS Asymptotics]** How is the LM (n-R-squared) statistic for testing q exclusion restrictions computed, and what distribution does it follow?

26. **[Part G: Ch.6 Multiple Regression Analysis: Further Issues]** If the dependent variable is divided by 1,000 (dollars → thousands of dollars), what happens to the OLS coefficients, standard errors, t statistics and R-squared?

27. **[Part G: Ch.6 Multiple Regression Analysis: Further Issues]** What is a beta (standardized) coefficient, and why is it useful?

28. **[Part G: Ch.6 Multiple Regression Analysis: Further Issues]** For rdintens = 2.613 + 0.00030 sales − 0.0000000070 sales^2 (sales in $ millions), give the turning-point formula for a quadratic and find where the marginal effect of sales becomes negative.

29. **[Part G: Ch.6 Multiple Regression Analysis: Further Issues]** In log(wage) = $\beta_0 + \beta_1$ educ + β_2 educ· pareduc + β_3 exper + β_4 tenure + u (pareduc = both parents' education), what is the return to one more year of education, and what does β_2 measure?

30. **[Part G: Ch.6 Multiple Regression Analysis: Further Issues]** Why is adjusted R-squared used, and when does adding a variable increase it?

31. **[Part G: Ch.6 Multiple Regression Analysis: Further Issues]** Why is a 95% prediction interval for one individual outcome y^0 much wider than the 95% confidence interval for $E(y|x = x^0)$?

32. **[Part G: Ch.6 Multiple Regression Analysis: Further Issues]** When the dependent variable is log(y), why is exp(predicted log(y)) not a good prediction of y, and how is it fixed?

Answers

1. [Part A: Econometrics 1–3 (Professor's Slides)] A population is the **entire** set of individuals, objects or data points we want to study, while a sample is a **subset** of the population that is **selected** for analysis.
Keywords: entire · subset · selected

2. [Part A: Econometrics 1–3 (Professor's Slides)] Studying the whole population is often too **costly** or **impossible**, so we study a sample; the goal is to find out what the **population is like** from the sample.
Keywords: costly · impossible · population is like

3. [Part A: Econometrics 1–3 (Professor's Slides)] External validity is the validity of applying the **conclusions** of a study **outside** the context of that study; it depends on whether the sample is **representative** of the population.
Keywords: outside · conclusions · representative

4. [Part A: Econometrics 1–3 (Professor's Slides)] β_1 is the change in log(price) from a **one-unit** increase in subsidy while **holding** the **other variables** (log(sunlight)) fixed, so it isolates the effect of subsidy alone.
Keywords: holding · other variables · one-unit

5. [Part A: Econometrics 1–3 (Professor's Slides)] Because battery and sunlight **interact**, the effect of battery on log(price) is $\beta_1 + \beta_3 \cdot$sunlight, which **depends** on the level of sunlight; the slope is **not constant**.
Keywords: interact · depends · not constant

6. [Part A: Econometrics 1–3 (Professor's Slides)] In random sampling every element of the population has an **equal probability** of being **selected**; in random assignment every element has an equal probability of being **assigned** to a group.
Keywords: equal probability · selected · assigned

7. [Part A: Econometrics 1–3 (Professor's Slides)] Ability is omitted, so it goes into the **error** u; because people with better ability get more education, educ is **correlated** with u, so $E(u \mid educ) = 0$ fails and β_1 is estimated with bias.
Keywords: ability · error · correlated

8. [Part A: Econometrics 1–3 (Professor's Slides)] A control variable is included to **hold** another factor **fixed** so it no longer sits in the error; adding ability helps satisfy the **zero conditional mean** condition and removes the **bias** in β_1.
Keywords: hold · zero conditional mean · bias

9. [Part B: Ch.1 The Nature of Econometrics and Economic Data] Ceteris paribus means "**other relevant factors** being **equal**"; most economic questions ask for the **causal** effect of one variable holding all other factors fixed.
Keywords: other factors · equal · causal

10. [Part B: Ch.1 The Nature of Econometrics and Economic Data] Panel data follow the **same** cross-sectional units **over time**; they can account for **time-invariant** unobservables and model lagged responses.
Keywords: same · over time · time-invariant

11. [Part C: Ch.2 The Simple Regression Model] OLS chooses the estimates that minimize the **sum of squared residuals**; a residual is the **actual** value y_i minus the **fitted** value $\hat{y}_i$.
Keywords: sum of squared residuals · actual · fitted

12. [Part C: Ch.2 The Simple Regression Model] $R^2 = SSE/SST$ is the **fraction** of the sample **variation** in y that is **explained** by the explanatory variable(s).
Keywords: fraction · variation · explained

13. [Part C: Ch.2 The Simple Regression Model] Homoskedasticity means the **error** u has the **same** (constant) **variance** for any value of x: $Var(u \mid x) = \sigma^2$.
Keywords: variance · error · same

14. [Part C: Ch.2 The Simple Regression Model] Unbiasedness means $E(\hat{\beta}_1) = \beta_1$: if sampling and estimation were **repeated** many times, the estimates would **on average** equal the **true** population value, though one estimate may differ.
Keywords: on average · repeated · true

15. [Part D: Ch.3 Multiple Regression Analysis: Estimation] There is no omitted variable bias if the omitted variable is **irrelevant** (its coefficient is zero) or **uncorrelated** with x_1.
Keywords: irrelevant · uncorrelated

16. [Part D: Ch.3 Multiple Regression Analysis: Estimation] Multicollinearity **increases** the sampling **variance** of the affected slope estimators, but OLS remains **unbiased**; it can be detected with variance inflation factors.
Keywords: variance · increase · unbiased

17. [Part D: Ch.3 Multiple Regression Analysis: Estimation] Under **MLR.1–MLR.5**, OLS is **BLUE** (best linear unbiased estimator): it has the **smallest variance** among all linear unbiased estimators.
Keywords: BLUE · smallest variance · MLR.1

18. [Part E: Ch.4 Multiple Regression Analysis: Inference] The p-value is the **smallest significance level** at which the null hypothesis would be **rejected**; we reject H_0 when the p-value is below the chosen level.
Keywords: smallest · significance level · reject

19. [Part E: Ch.4 Multiple Regression Analysis: Inference] Not necessarily: statistical significance only shows the effect differs from zero, while **economic** (practical) significance depends on the **magnitude** of the coefficient.
Keywords: not necessarily · magnitude · economic

20. [Part E: Ch.4 Multiple Regression Analysis: Inference] In **repeated** samples, intervals built this way **contain** the **true** population coefficient in 95% of cases; the bounds are random, not the parameter.
Keywords: repeated · contain · true

21. [Part E: Ch.4 Multiple Regression Analysis: Inference] Heteroskedasticity (i) and an **omitted** important variable (iii) violate the CLM assumptions, so the t statistics are invalid; a high **correlation** of 0.95 (ii) is not a CLM violation (only perfect collinearity is ruled out).
Keywords: heteroskedasticity · omitted · correlation

22. [Part F: Ch.5 Multiple Regression Analysis: OLS Asymptotics] An estimator is consistent if, as the **sample size** grows, its distribution **collapses** to the **true** parameter (plim $\hat{\beta}_j = \beta_j$); OLS is consistent under **MLR.1–MLR.4** (even zero correlation between each x_j and u is enough).
Keywords: sample size · collapse · true · MLR.4

23. [Part F: Ch.5 Multiple Regression Analysis: OLS Asymptotics] The inconsistency is **positive**: plim $\tilde{\beta}_1 - \beta_1 = \beta_2\delta_1$, with $\beta_2 > 0$ (higher **risktol** raises pctstck) and $\delta_1 > 0$ because funds and risktol are positively **correlated**; it does **not disappear** as n grows.
Keywords: positive · risktol · correlated · not disappear

24. [Part F: Ch.5 Multiple Regression Analysis: OLS Asymptotics] No: most adults smoke **zero** cigarettes and cigs is **nonnegative** and discrete, so it is **not normal**; with a **large sample**, asymptotic normality still makes the t and F tests approximately valid.
Keywords: zero · nonnegative · not normal · large sample

25. [Part F: Ch.5 Multiple Regression Analysis: OLS Asymptotics] Estimate the **restricted** model, regress its **residuals** on all the explanatory variables, and compute LM = $\mathbf{nR^2}$ from that auxiliary regression; under H_0 it is approximately **chi-square** with q degrees of freedom.
Keywords: restricted · residuals · nR · chi-square

26. [Part G: Ch.6 Multiple Regression Analysis: Further Issues] All coefficients and standard errors are **divided by 1,000**, while the **t statistics** and **R-squared** stay **unchanged**; only the units change, not the statistical conclusions.
Keywords: divided · unchanged · t statistic · R-squared

27. [Part G: Ch.6 Multiple Regression Analysis: Further Issues] A beta coefficient comes from regressing the **standardized** y on standardized x's; it is the change in y, in **standard deviations**, from a one standard deviation increase in x_j, so effects measured in different **units** can be **compared**.
Keywords: standardized · standard deviation · units · compare

28. [Part G: Ch.6 Multiple Regression Analysis: Further Issues] Turning point x^* = $\mathbf{\hat{\beta}_1/(2|\hat{\beta}_2|)}$ = 0.00030/(2 × 0.0000000070) ≈ **21,428.57**, i.e. sales of about **$21.4 billion**; beyond that the estimated marginal effect of sales on rdintens is negative.

29. [Part G: Ch.6 Multiple Regression Analysis: Further Issues] The return is $\beta_1 + \beta_2 \cdot$pareduc, so it **depends** on **parents' education**; the **interaction** coefficient β_2 is how much the return changes for one more year of pareduc (estimate 0.00078: at pareduc = 32 the return is about 7.2%).
Keywords: depends · parents · interaction

30. [Part G: Ch.6 Multiple Regression Analysis: Further Issues] R-squared **never falls** when a variable is added, so adjusted R-squared imposes a **penalty** for extra regressors using **degrees of freedom**; adding a variable raises it only if that variable's **t statistic** exceeds one in absolute value.
Keywords: never falls · penalty · degrees of freedom · t statistic

31. [Part G: Ch.6 Multiple Regression Analysis: Further Issues] The confidence interval reflects only the sampling **variance** of $\hat{y}^0$, while the prediction interval also includes the variance of the unobserved **error** for an **individual** outcome (σ^2), which does not shrink as n grows (e.g. colgpa: CI 2.66–2.74 vs PI 1.60–3.80).
Keywords: error · variance · individual

32. [Part G: Ch.6 Multiple Regression Analysis: Further Issues] exp(log $\hat{y}$) systematically **underestimates** E(y|x); we **multiply** it by $\hat{\alpha}_0 = \exp(\hat{\sigma}^2/2)$ under **normal** errors, or by the average of $\exp(\hat{u}_i)$ from the residuals.
Keywords: underestimate · multiply · residuals